



SIGs Table Instructions

This document was last updated on May 19, 2025.

Regimens are the core concept class of HemOnc. The purpose of this document is to describe the various fields in the “sigs” table, which captures the granular details of regimens and their variants. It will be expanded over time.

Variables in SIGs table

The sigs table has a somewhat complex data structure, with four general datatypes:

- 1) **Anchors** – variables that serve to connect this table to other tables in the KB, but do not otherwise have a definitional quality.
- 2) **Binders** – variables that serve to connect one or more rows of the sigs table, within the context of a given regimen variant. Some of these are ordered, and some of them are not.
- 3) **String variables** – variables that define the unique SIG for a given regimen variant, in a character-based format. These exist to aid human readability, and, in some cases, there is no CUI counterpart.
- 4) **CUI variables** – variables that define the unique SIG for a given regimen variant and that have been formalized as OMOP-compatible concepts, with unique CUIs.
- 5) **Intermediate variables** – these variables are not definitional but not metadata either; they are typically an intermediate form parsed from the HemOnc.org website and are left within the table to aid human readability
- 6) **Metadata**.

The following is a list of variables that have CUI counterparts, which are designated by the same name, appended with “_cui”:

String-based variable	CUI-based variable
regimen	regimen_cui
component	component_cui
doseUnit	doseUnit_cui
doseCapUnit	doseCapUnit_cui
route	route_cui
durationUnit	durationUnit_cui
frequency	frequency_cui
variant ¹	variant_cui

In addition to these datatypes, some of the variables in the table can take on multiple values:

Variable	Datatype	Example	Reason
study	anchor	Study 1 Study 2	More than one study can provide supporting evidence for a given regimen variant



timing_sequence	string	1,2,3,4	Many SIGs are repeated at a cyclic interval, e.g., every 21 days for 4 cycles
allDays	string	1,8,15,22	Many SIGs are repeated more than one time within a cycle or course, e.g., every 7 days for 28 days
seq.rel.when	string	-14,-7	Some drugs are given more than once relative to a second drug, e.g., once 14 hours before and once 7 hours before

General structure with cardinality:

Variant captures the regimen variant (specified by **variant_cui**)

Variant contains 0 or more portions (specified by **portion**)

Portion contains 1 or more components (specified by **component/component_cui**)

Component has 1 or more steps (specified by **step_number**)

Step is repeated once or more than once (specified by **cycle_length_lb**, **cycle_length_ub**, **cycle_length_unit**, **timing_sequence**)

Component is administered once or more than once within a repeat (specified by **allDays**)

Component is administered once or more than once within a day (specified by **frequency/frequency_cui**)

Supporting Studies

The first variable in the sigs table represents the study (or studies) that was/were used as evidence for a table entry. The vast majority of these studies are prospective clinical trials; a small minority are case series or retrospective studies. A very small number of entries have no study support at all; for example, a case where the FDA makes a dose recommendation that was not explicitly described in the supporting study. As the HemOnc KB grows, the supporting studies for a given regimen may also grow; it is important to note that the study variable, while acting as an anchor, does not in itself define any of the rest of the SIG. Here is the row-wise distribution of studies (as of October 2024):

Number of supporting studies	Rows (% total)
0	70 (0.3%)
1	17,197 (74.4%)
2	3639 (15.7%)
3	997 (4.3%)
4	398 (1.7%)
5	263 (1.1%)
6 or more	553 (2.4%)

It should be noted that the number of rows with single-study support is modestly inflated, due to the fact that some single studies have a large number of table entries (e.g., a COG protocol). This becomes clear when summarizing by study uniqueness:



Unique supporting studies	N (%)
Singlet	2989 (65%)
Doublet	844 (18%)
Triplet	293 (7%)
Quadruplet or more	481 (12%)
Total	4607 (100%)

Components and Roles

All SIGs in the sigs table are comprised of one or more components.

component_role: Each component can take on one of three roles: 1) *primary systemic* – drugs that distribute throughout the body and have the primary intent of modifying the underlying disease state, whether through direct (e.g., cytotoxic chemotherapy) or indirect (e.g., immunotherapy) means; 2) *secondary systemic* – drugs that distribute throughout the body with the primary intent of ameliorating one or more side effects introduced by primary systemic or locoregional components; 3) *locoregional* – drugs or other interventions that have a localized effect (e.g., surgery, intralesional injections, external beam radiotherapy).

SIG variables

All SIGs in the sigs table have a **class** and one or more **parts**.

class: This variable describes whether the SIG (or part of a multipart SIG) takes on a normal (“canonical”) form, is a radiation SIG, or has a non-canonical form. There are three types of canonical forms:

IV intermittent canonical Sig is a SIG for an intravenous medication that has at least the following information: component, dose, dose unit, route, frequency, schedule.

Example 1: Cisplatin 70 mg/m² IV once on day 1

Example 1: [Component] [Dose] [Dose Unit] [Route] [Frequency] [Schedule]

Note that these SIGs can have additional information; this is the minimum requirement.

Example 2: Cisplatin 70 mg/m² (maximum dose of 140 mg) IV over 30 to 60 minutes once on day 1, given second

Example 2: [Component] [Dose] [Dose Unit] ([Dose Cap] [Dose Cap Unit]) [Route] [Duration Min] [Duration Max] [Duration Unit] [Frequency] [Schedule] [Sequence]



Non-IV canonical Sig is a SIG for any medication other than those given by the intravenous route that has at least the following information (the same requirements as *IV intermittent canonical Sig*): component, dose, dose unit, route, frequency, schedule.

part: Every SIG has at least one part, and a number of SIGs have multiple parts. Note that each part of a SIG is assigned a class, such that a single multi-part SIG can have different classes. For example, most prednisone tapers specified in protocols have defined (canonical) parts, and a final undefined (non-canonical) part. Here is an example:

- Prednisone 30 mg PO once per day on days 1 to 7, then 20 mg PO once per day on days 8 to 14, then 10 mg PO once per day on days 15 to 21, then tapered off over the next 14 days.
- [Component] [Part 1 of 4] [Part 2 of 4] [Part 3 of 4] [Part 4 of 4]

In this example, there are four parts

Cycle/course timing variables

timing_sequence: This variable captures timing for both non-repeating and repeating (cyclic) regimens. There are two major types of cyclic regimens: definite and indefinite. The timing sequence variable specifies the exact cycles in which a given component of a regimen is delivered. The values follow the following definitions:

Standard form for non-repeating SIGs: a single integer. For example, 1 means that the component is given in the first (and usually only) course. Some regimens are composed of sequential courses, such that a particular SIG in that regimen may have a cardinality of greater than 1.

Standard form for repeating (cyclic) SIGs: a comma-separated list of integers. For example, 1,2,3,4 means that the component is given in the first four cycles of a regimen, whereas 2,4,6,8 means that the component is given in the 2nd, 4th, 6th, and 8th cycles of a regimen. The default assumption is that the first cycle of therapy is cycle 1, but there are some exceptions, e.g., it is common to refer to a pre-phase treatment as cycle 0.

Optional cycles with a specified end: if a component can be given but does not have to be given, we use parentheses to denote this. For example, 1,2,(3),(4) denotes that a component must be given in the first 2 cycles and is then optional for the 3rd and 4th cycles.

Optional cycles without a specified end: if a component can be given optionally after some required period, without an ending specified, we use a "+n" in parentheses to denote this. For example, 1,2,3,4,(+n) denotes that a component must be given for at least the first 4 cycles and then can be optionally given in subsequent n cycles. An example: 1,(+1) means that the component must be given for the first cycle and then can optionally be extended one cycle at a time (to cycle 2, 3, and so forth); this is a common nomenclature used for cycleSIGs that are indeterminate, e.g., "21-day cycles". In some very unusual cases, the repeat is a skip repeat,



e.g., “Odd cycles”. In these cases, the repeat is denoted by **1,(+2)**, meaning that the component is given in the 1st, 3rd, 5th, etc. cycle. *As of May 19, 2025 there are 4208/23832 (18%) rows with the (+1) notation, and 15/23832 (0.06%) with the (+2) notation.*

Optional cycles with a conditional end: if a component can be given optionally until some predefined criteria has been met (e.g., given until complete remission), we use a “+c” in parentheses to denote this. For example, **1,2,3,4,(+c)** denotes that a component must be given for at least the first 4 cycles and then can be optionally given in subsequent cycles until a conditional criterion is met. Right now, the conditional criteria will be described in the `cyclesigs_note` field; in the future, we will likely develop a dedicated variable for conditional endings.

timing: This variable takes on a value if a component has a declared dosing schema for one or more differing time points of a given regimen. On HemOnc.org (screenshot below), these types of instructions are delineated by the “as follows:” instruction, with the cycle- or time-specific instructions below. For example, in the *Carboplatin, Pegylated liposomal doxorubicin, Bevacizumab* regimen (regimen CUI 5704), variant CUI 130497, bevacizumab is given by two different schedules. In cycles 1 to 6, the dosing is 10 mg/kg on days 1 and 15; from cycle 7 onwards, the dosing is 15 mg/kg on day 1.

Targeted therapy [[edit](#) | [edit source](#)]

- **Bevacizumab (Avastin)** as follows:
 - Cycles 1 to 6: 10 mg/kg IV once per day on days 1 & 15
 - Cycle 7 onwards: 15 mg/kg IV once on day 1

28-day cycle for 6 cycles, then 21-day cycles

Cycle/course length variables

Cycle (or course) length is captured by two magnitudes, **cycle_length_lb** and **cycle_length_ub**, paired with a unit variable, **cycle_length_unit**. Cycle length variability is not uncommon, for example a cycle might be expressed as “21- to 28-day cycles” in which case, `cycle_length_lb` would have a value of “21”; `cycle_length_ub` would have a value of “28”; and `cycle_length_unit` would have a value of “day”. There are several special cases to consider:

Courses with a definite lower bound, indefinite upper bound: For example, “at least 6 months”. In this case, `cycle_length_ub` is assigned the value “NUB” (No Upper Bound). Note that this can only apply to courses; cycles by their nature must have an end in order to be cyclic.



Cycles (or courses) without enumerated values: These are quite common, for example, “One course” is a frequently encountered cycleSIG. In this case, `cycle_length_lb` and `cycle_length_ub` are both assigned the value of “1” and `cycle_length_unit` is assigned the value of “indeterminate”.

Conditional length of treatment: There are some cases where the lower and/or upper bounds may be conditional. For example, “Continued until CR or 60 days”, which means that the medication is continued until complete remission and/or 60 days have elapsed. In this case, `cycle_length_lb` is assigned “c” for conditional, and `cycle_length_ub` is assigned “60”; `cycle_length_unit` is assigned “day”. At this time, we do not explicitly model the conditionality, although this may change in the future.

Note that for convenience in modeling we make several transformations of lengthy or unspecified courses, as follows:

- Indefinite courses (e.g., a drug taken once per day and continued indefinitely) are transformed into 28-day cycles
- 12-month courses are transformed into 28-day cycle x 13 cycles
- 2-year courses are transformed into 28-day cycle x 26 cycles
- 3-year courses are transformed into 28-day cycle x 39 cycles
- and so forth for 4-year, 5-year, and so on

tail: This variable captures portions of SIGs that don’t fit into any other category. This includes generic statements about the total dose of a given drug per cycle, etc. Because of the way in which blobs are constructed (see below), the tail variable must always take on a value, the default value being “EMPTY”.

SIG scheduling and frequency variables

The following variables define the schedule of administration at the component level within a given cycle or course of therapy: **allDays**, **frequency/frequency_cui**, and **divided**.

allDays

The **allDays** variable captures the exact days on which a component is administered, using two different conventions:

Convention 1: For relatively simple and/or discontinuous scheduling, the days are represented by a comma-separated vector, e.g., 1, 2, 3, 4. In addition to explicit scheduling, there are a few special cases, similar to the `timing_sequence` variable:



Negative days are padded with a caret, to prevent auto-formatting issues when importing the CSV into programs such as Microsoft Excel. For example, day -1 is represented as “^1”. Whenever possible, sequences that start with a negative day are expressed using convention 2 (see below).

Optional days are enumerated with parentheses, e.g., “1,3,(5)” means the doses are days 1 and 3 are required, and the dose on day 5 is optional.

Indefinite optional extensions are represented by “(+n)” where n is the number of days, e.g., “1,(+1)” means that doses are days 1,(2),(3),(4),...

Conditional endings are represented by “(+c)”; these are uncommon and difficult to completely model at this time.

Choice of day is represented by the OR logical (vertical pipe). For example, “7|8|9” means that the component can be administered on day 7 or day 8 or day 9.

Unspecified day is represented by “U” for a single day, and “U1,U2,U3,...” for multiple unspecified days. Note that this nomenclature is only used when there is explicit declaration that the day(s) is/are unspecified; if the day is missing on HemOnc.org, the field is left blank.

Convention 2: For lengthier schedules that can be expressed by a regular repeating interval such as one (1) day, a compact representation is used with this format: “[from, to, by]”. For example, “[1,7,1]” is the same as “1,2,3,4,5,6,7”, and “[(2),(4),1]” is the same as “(2),(3),(4)”. This format also easily captures alternate-day scheduling, for example, “[1,14,2]” is the same as “1,3,5,7,9,11,13”. However, it cannot capture staggered schedules such as three days per week (days 1, 3, 5). Nor can it capture vectors that are a hybrid of required days and optional days.

Note that the two conventions can be combined, e.g., a drug that is given for 56 sequential days and then optional extended indefinitely beyond this initial period can be represented as “[1,56,1],(+1)”. Discontinuous schedules can be concatenated such as this example: “[1,7,1],[15,21,1],[29,35,1]”. A special case of combination would be a regimen with a defined duration represented by convention 2, followed by a condition, e.g., “[1,90,1],(+c)”. This particular representation means that the drugs is continued for a maximum of 90 days *or until a condition is met*; note that, as stated above, conditional endings are not further modeled at this time due to their significant complexity.

Additional notes

First day conventions

While most SACT regimens begin on day 1 by convention, there are exceptions to this. Some pediatric regimens begin on day 0, and transplant/biologic regimens typically begin with a negative number relative to the day of transplant, which is day 0. Procedures are also conventionally coded as day 0, such that the first day after a surgery is post-operative day 1 (POD1).



Inclusion of day 0 conventions

For those regimens that have negative days, there is an additional complexity. Transplant and biologic therapy regimens include day 0 (e.g., days -2, -1, 0, +1, +2, etc) whereas most other regimens do not recognize a day 0 (e.g., days -2, -1, +1, +2, etc).

Timing of drugs with a weekly schedule

Treatments given on a weekly basis present some additional difficulties when expressing specific days in the **allDays** variable. For example, many SACT trials that use a repeating weekly dosing schedule express the timing as “once per week”, “twice per week”, or “three times per week” without specifying exact days. And many trials that utilize radiation therapy express the timing as “five times weekly”. For these cases, we use the following conventions, which are guided by an assumption that treatment starts on a Monday:

SIG string	allDays	Day(s) of week
Once per week	1	Monday
Twice per week	1,4	Monday, Thursday
Three times per week	1,3,5	Monday, Wednesday, Friday
Five times per week	[1,5,1]	Monday thru Friday

Timing of daily drugs given on a monthly or yearly schedule

This is also commonly encountered in the literature and presents the additional difficulty that months have differing time periods. We adopt the convention that a monthly period equals 30 days, and that a year, also expressed as twelve (12) months, equals 365 days. For those rare instances where a total duration is something like eighteen (18) months, we concatenate a year-based duration with a month-based duration in **allDays**, as shown in the table:

Duration	allDays
One (1) month	[1,30,1]
Two (2) months	[1,60,1]
Three (3) months	[1,90,1]
Four (4) months	[1,120,1]
Six (6) months	[1,180,1]
Twelve (12) months aka one year	[1,365,1]
Eighteen (18) months	[1,365,1],[366,545,1]



Two years	[1,730,1]
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frequency

The **frequency/frequency_cui** variables specify the scheduling of medication administration on a given calendar day, with one important exception. The following table provides the currently supported frequency values:

Frequency of dosing	Allowed values [<i>conventional acronym</i> ¹]
One-time	once; twice
Once daily	once every evening [<i>QHS</i>]; once every morning [<i>QAM</i>]; once per day [<i>QD</i>]
Two (2) times daily	every 12 hours ² ; twice per day [<i>BID</i>]
Three (3) times daily	every 8 hours ² ; three times per day [<i>TID</i>]
Four (4) times daily	every 6 hours ² ; four times per day [<i>QID</i>]
Five (5) times daily	five times per day
Six (6) times daily	every 4 hours ²
Frequency not specified	daily NOS

¹Note that we follow JCAHO conventions and avoid the use of these acronyms due to the potential for confusion between some of them. Also, QOD (every other day) does not exist in our model as every-other-day dosing is captured by the **allDays** variable.

²Conventions for exact timing vary between institutions, e.g., every 12 hours might be 0600 and 1800 at one institution, and 0800 and 2000 at another; we do not further specify the hour(s) of administration for this reason.

The important exception are those drugs that are given in a continuous infusion that crosses more than one calendar day. These are marked as frequency of “continuous” and the infusion time is specified in the duration variables (to be completed).

divided

The **divided** variable represents the uncommon scenario where a total daily dose is provided in a reference, with an explicit instruction to split the dose but without further enumeration. For example, “prednisone 60 mg/m²/day PO in divided doses” would be represented by frequency = “daily NOS” and divided = “TRUE”, since it is likely that the division of dose would not be even (e.g., more might be given in the mornings than in the evenings). If a reference does specify that the division is into equal daily doses, we prefer to avoid using divided. In the preceding example, if the drug were divided into two equal daily doses, we would represent it as dose = 30 mg, frequency = “twice per day”, and divided = “FALSE”.

Duration Variables



Duration variables are used to specify the length of an infusion.

Certain durations are expressed as words without explicit time definitions, e.g., “push”. For these word-based durations, we have adopted standard timings based on input from pharmacist members of our editorial board, as follows:

Phrase	Duration in HemOncKB
rapid infusion	1 to 2 minutes
rapid injection	1 to 2 minutes
bolus	1 to 30 minutes
short infusion	3 to 15 minutes
push	3 to 15 minutes
slow push	16 to 29 minutes

Relative timing variables

This set of variables specifies the administration of components of a regimen *relative to other components* of the regimen.

Regimen Variants

Unique Regimen Variant Representation - Initial

A unique regimen variant is instantiated once a **variant_cui** value has been assigned. All rows in the sigs table with a particular variant_cui comprise the entirety of the *necessary parts* of the regimen. Without exception, any component with a component role of primary systemic is a necessary part.

In order to create a unique regimen variant representation for a regimen variant that is not yet assigned a CUI (i.e., the **variant_cui** field has a value of TBA¹), two steps are undertaken:

¹ This is a manual value assignment to rows of a regimen variant with a primary systemic component, a radiation therapy component, or occasionally a local therapy component if it is deemed critical to the definition of the regimen. It will not usually be present in the published HemOnc products.



- 1) The SIGs table is ordered by 1) **regimen**, then by 2) **variant**, then by 3) **portion**, then by 4) **component**, then by 5) **timing**, then by 6) **step_number**
- 2) For each unique regimen-variant combination, a pipe-delimited blob is created by concatenating four parts:
 - a. Header: this value appears once per blob, i.e., it does not repeat
 - b. Body top: this value can be repeated, if there are more than one row of data present to represent the variant, in the order in which they are arranged by step 1.
 - c. Body middle: this value can be repeated, if there are more than one row of data present to represent the variant, in the order in which they are arranged by step 1.
 - d. Body bottom: this value can be repeated, if there are more than one row of data present to represent the variant, in the order in which they are arranged by step 1.

The resulting blob is then compared against all valid blobs in the variant.table (i.e., those with the variable valid = TRUE). If there is no match, a new regimen variant CUI is created and the CUI is added to the **variant_cui** field in sigs; a new entry in variant.table is also created.

The variant blob captures the discrete variables in a string-based representation.

Section	V1	V2
Header	regimen	
Body top	portion	
Body middle	component, component_role, timing, branch, branch_type, cyclesigs, step_number, doseMinNum, doseMaxNum, doseUnit, divided, doseCapNum, doseCapUnit, route, allDays, durationMinNum, durationMaxNum, durationUnit, frequency, inParens, sequence, seq.rel, seq.rel.what	
Body bottom	tail	

Unique Regimen Variant Representation - Updates

It is often the case that an existing regimen variant undergoes slight modifications, such as the addition of an infusion timing instruction for one or more drugs, or a correction to a detail that does not fundamentally change the regimen variant. In these cases, an update is created by undertaking the same blob creation procedure as above for each variant with a **variant_cui** value. If the newly created blob does not match that in variant.table for the version 1 of the



regimen variant, a new version 1 is created, and all prior versions are decremented by 1 and marked as invalid (i.e., the variable valid = FALSE).

Additional Notes

Continuous oral medications: These pose a significant problem because the cycle length is arbitrary, as the medications are typically taken every day (or every other day). On HemOnc.org, the typical instruction for a continuous medication without a definite end date is “**Continued indefinitely**”. Conversely, defined courses of treatment have been expressed in different ways, often depending on whether the original publication explicitly defined a cycle length. In reality, all continuous administrations are subject to refill quantities and periodic re-evaluations by the treating clinician. Given that the typical refill interval is monthly and given that the most commonly expressed cycle length is 28 days (4 weeks), we have moved to adopt a universal 28-day cycle standard for continuous oral medications, unless the total length of treatment is less than 28 days or is not easily expressed in multiples of 28.

Required secondary systemic medications: A subset of secondary systemic medications, while not comprising the core components of a systemic anticancer treatment regimen, are nevertheless required. When this is the case, the secondary systemic medication will be labeled with the variant_cui; when there is optionality, they are instead labeled as “N/A - secondary systemic drug”. Please note that because of this requirement, a secondary systemic medication could cause the creation of a new regimen variant. For example, the regimen “Abiraterone monotherapy” has a single dosing for the primary systemic drug abiraterone but must be given with either prednisone *or* prednisolone. The following is an evolving list of such examples:

Autologous stem cells: When used, they are always required.

Mesna: When used, it is always required.

Prednisolone: When used with abiraterone, it is required.

Prednisone: When used with abiraterone, it is required.

Appendix

Transform compact allDays to full vector format

Placeholder for R script to transform convention 2 (compact) allDays representation to fully expressed vector format.